

Linear Algebra Notes - Eigenvalues

Course: MATH210 - Linear Algebra

Key idea

For a square matrix A , a nonzero vector v is an eigenvector if $A v = \lambda v$ for some scalar λ . The scalar λ is the corresponding eigenvalue.

Procedure

1. Compute $\det(A - \lambda I) = 0$.
2. Solve the characteristic polynomial for λ .
3. For each eigenvalue, solve $(A - \lambda I)v = 0$.
4. Check whether enough independent eigenvectors exist for diagonalization.

Quick practice

For $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, identify the eigenvalues and one eigenvector for each. Then explain why A is already diagonal.